

ECON 205: Statistics for Business and Social Sciences

University of Northern British Columbia

FALL 2024

1 Contact Information

Instructor: Leandro Freylejer

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Office Hours: Tuesdays 12:00pm-1:00pm, Fridays 1:20pm-2:20pm

2 Course Information

Room and Time: Teaching Lab 8-166 , **Time:** Mondays, Fridays 2:30pm-3:50pm

Course website: <https://moodle.unbc.ca/>

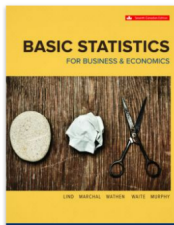
Teaching/Lab Assistant, Office Hours: TBA

The TA for this course is Gabby Teye-Ali (email:teyeali@unbc.ca). Gabby is an excellent MSc in Business Administration student who will teach labs as well as grade assignments, problem sets and exams. Do not hesitate to email him should you have any questions regarding lab assignments or problem sets.

Overview

This course provides an introduction to the principles and applications of statistics relevant to the social and management sciences. The course is divided into four core sections. In the first section, students learn to use descriptive statistics to organize and analyse data. In the second section, students are exposed to several topics in probability theory. The last two sections teach students several techniques for using observational data to infer properties of an underlying population (statistical inference). In addition to lectures, students are required to complete lab sessions. In these laboratories, students will learn to use the statistical software MINITAB. No previous computer experience is required.

3 Textbook



The **required textbook** for this course is *Basic Statistics for Business and Economics*, by Lind, Marchal, Wathen, Waite, and Murphy 7th Canadian ed, McGraw-Hill (5th and 6th editions are fine)

4 Lectures, Tutorials, Labs and Office Hours

Lectures

Lectures will take place on Mondays and Fridays from 2:30pm to 3:50pm in Teaching Lab 8-166. In most weeks, I will post lecture slides on the course website at least one and a half hours before class. If necessary, corrected slides will be posted as soon as possible. Slides are not substitutes but complements to class attendance. I also encourage students to complete readings before class. This facilitates in-class discussion and participation. I encourage you to use laptops or tablets only for the purposes of taking notes. I ask you to please turn off your cell phones and other communications devices before entering the classroom. Please be mindful of other students around you. Any conduct that is deemed to be disruptive to a learning environment will not be tolerated.

Tutorials

There will be two tutorials in this class scheduled for October 21st and December 2nd during class time. In these tutorials, I will go over solutions to assignments and additional problems, which will be posted on Moodle prior to the tutorial. It is important that you attend tutorials and come prepared to ask questions. Tutorials are scheduled for the class immediately preceding exams, and therefore, they might be your last chance to ask questions before being tested on the material.

Labs

There will be a total of five labs. Labs are scheduled for the weeks of September 30th to October 4th, October 14th to October 18th, October 28th to November 1st, November 11th to November 15th, and November 25th to November 29th. The exact time and location of each lab section can be found at the UNBC website. During the labs, the lab instructor will go over computer applications of the statistical techniques I present in class. All lab assignments will be posted on the course website in advance.

Lab Assignments are due before midnight on the **Friday** of the week following the in-person lab (this is irrespective of the day of the week, in which students are scheduled to attend the lab). All labs must be submitted on-line through Moodle.

Office Hours

Students are also encouraged to attend my office hours. A lot of the later material in this course builds on earlier material. Therefore, it is very important that you do not let major gaps in your understanding drag on. You are strongly encouraged to come to office hours as early as possible and as often as you need.

If for some specific reason you cannot attend office hours, I am available by appointment only. Please do not drop-in unannounced.

5 Evaluation

Due Dates and Final Grade Breakdown

The final grade for the course consists of **term work**, worth 64%, and a **comprehensive final exam**, worth 36%. The term work is composed of four problem sets, worth 6% each, five lab assignments, worth 4% each, and a midterm exam, worth 20%. The following are the due dates for graded work (**see above for lab assignments**).

	Weight	Due date
First Problem Set	6%	October 4th before midnight
Second Problem Set	6%	October 18th before midnight
Midterm Exam	20%	October 25th in-class
Third Problem Set	6%	November 15th before midnight
Fourth Problem Set	6%	November 29th before midnight
Final Exam	36%	TBD

Throughout the semester, I might assign additional problems and/or readings. These will not be graded but I expect that you attempt to solve problems and go over readings before class.

Problem Sets and Lab Assignments

There will be four graded **problem sets** in this course. In these problem sets, I will ask you to solve problems similar to the types of problems that will appear on exams. I will post these problem sets on the course website *at least* a week before the due date. I encourage you to start working on them as soon as they are posted.

In addition to problem sets, there will be five labs due on the dates indicated above.

Problem sets and lab assignments should be submitted through Moodle before midnight on the due date. You can either upload typed solutions or scans/photos of handwritten answers to Moodle. Please, whatever method you choose, make sure the uploaded document is legible. The TA will grade your problem set/assignment in Moodle and you will be able to see your score, as well as

your mistakes, on-line. **I do not accept e-mail submissions.**

Late problem sets and lab assignments are subjected to a penalty of 8 percentage points per day, and they will not be accepted two days after the deadline has passed (including weekends). If you cannot submit an assignment on time, for an exceptional reason out of your control, you should email me as soon as you are able to. We will discuss your particular situation, and hopefully, come up with a mutually satisfactory solution, which is also fair to other students in the class who submitted their assignment on time. When proper documentation can be obtained, you should be prepared to provide such documentation.

Exams

There will be two exams in this class. A midterm exam scheduled for Friday, October 25th during class time, and a 3-hour **comprehensive** final exam to be scheduled by the University during the examination period. I will provide you with more information regarding the format of these exams closer to the examination dates.

6 Administrative Details

Course web-page and e-mail policy

Students should check Moodle (moodle.unbc.ca) frequently for announcements, assignments, problem sets, additional readings and other information for this course. Moreover, I often post additional practice material, such as past exams with answer keys, before midterm and final exams.

I will do my best to respond to e-mails within 24 hours on a weekday, 48 hours on a weekend. Please use official “@unbc.ca” email accounts for all communication. I only respond to e-mails posing questions that can be answered in no more than three sentences. For more involved questions, I will ask you to attend my office hours or ask questions during lectures. Also, I do not reply to e-mails that request information that can be found in the website or the syllabus.

Academic Misconduct

I do not tolerate any form of academic misconduct, including *plagiarism*. Any student caught engaging in such activities will be subject to academic discipline ranging from a mark of zero on the test or examination to dismissal from the university as outlined in the academic handbook.

Missing an Exam

A missed midterm or final exam will receive a zero grade, unless in exceptional circumstances when for documented serious reasons a student is not able to write an exam. In this case, I usually shift the weight of the midterm towards the final exam or allow a student to write a make-up final exam. If you miss an exam, you should email me as soon as you are able to. We will discuss your particular situation, and hopefully, come up with mutually satisfactory solution, which is also fair

to other students in the class who did not missed the exam. In the case when proper documentation can be obtained, you should be prepared to provide such documentation.

Grade Appeals

Appeals regarding the grading of assignments or problem sets must be submitted to me by email within two weeks of your receipt of the graded work. You must clearly and in detail explain the reason for your appeal. I will re-grade the entire work. Note that this may lead to a higher or lower overall grade.

7 Tentative schedule of Topics and required readings

Section	Topic	Chapter
Introduction	Introduction to Statistics	1
Descriptive Statistics	Organization and presentation of data	2
	Numerical methods for describing data	3
Probability Theory	Introduction to probability	4
	Discrete Probability Distributions	5
	Continuous Probability Distributions	6
Statistical Inference	Sampling Methods and CLT	7
	Estimation and Confidence Intervals	8
	Hypothesis Testing	9, 10,+
	Analysis of Variance	11
Topics in Statistics	Hypothesis Testing for Multiple Nominal Categories	14

+ indicates that additional notes will be posted in the “Additional Material” folder in Moodle.

8 Additional University Resources

Academic Success Centre

The Academic Success Centre provides students with FREE access to academic support services:

- Tutoring (by appointment, asynchronous online, or drop-in)
- Personalized study skills assessments
- Peer-led course supports
- Downloadable handouts
- Access to self-assessment sites

- Customized programs and workshops

ASC services are available in person at the Prince George Campus and online for distance, regional, or physically distanced students.

For more information see the Academic Success Centre website at www.unbc.ca/asc, to book an appointment visit them on the first floor of the library, email asc@unbc.ca, or phone 250-960-6367. Toll-free: 1-888-440-3440 (wait on the line and ask for the ASC).

Access Resource Centre

The Access Resource Centre (ARC) provides service to students with documented disabilities or health conditions, ranging from permanent to temporary, including but not limited to chronic health issues, hearing and visual impairments, learning disabilities and attention deficit/hyperactivity disorders, mental health and neurological disabilities, and mobility and other physical disabilities. ARC staff are available by appointment to assess specific needs, provide referrals, and arrange appropriate academic accommodations to assist students in achieving their academic goals. Students who may have a need for academic accommodation are encouraged to contact ARC by email at arc@unbc.ca, by phone at 250-960-5682 (toll free 1-888-960-5682), or in person at 5-157.

Health and Wellness

Everyone can experience mental health concerns or stressful events that can interfere with our academic performance and negatively impact our daily activities. All of us can benefit from support during times of struggle. If you or anyone you know experiences academic stress, difficult life events or feelings of anxiety, depression or suicidal thoughts, the UNBC Wellness Centre team is here to help. Their services are free for UNBC Students and appointments are available. You can learn more about confidential mental health services available on and off campus at: <https://www2.unbc.ca/counselling> OR <https://www2.unbc.ca/medical-clinic>. Remember that getting help is a smart and courageous thing to do - for yourself, for those you care about, and for those who care about you. Asking for support sooner rather than later is almost always helpful.